Chatterbots with Occupation – Between Non Task and Task Oriented Conversational Agents

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Abstract. Chatterbots are computer systems specialized in simulating intelligent conversation. Dialog systems are often classified as task and non-task oriented: one focusing on the completion of particular tasks, and the other representing the less formal aspects of interaction with an system, such as free talking. In our research we aim to create a performing conversational system that would act as a language tutor. Performing such function means not only passing knowledge, but also assisting students in developing a variety of other attributes, as well as creating a necessary teacher – student bond. In this paper we would like to present an idea of a conversational system that would perform both tasks, reaching beyond the boundaries of existing dialog systems classification. First, we discuss the aspects of currently used chatterbot systems with a certain purpose. Then we address the existing problems of maintaining conversation with users and reacting to their emotional states in tutoring systems. We also present the current work on our tutoring system as an example of chatterbot with occupation. Finally we discuss the potential applications of such chatterbots for education and other fields.

1. INTRODUCTION

In 1950 Alan Turing started his paper with famous question: "Can machines think?" [28]. Soon after, the possibility for machines to think started to be explored by a newly founded scientific field of Artificial Intelligence (AI). A number of technologies that tried to attempt answering Turing's question have emerged in Computer Science over the following years. Diaogue systems are the example of technology that took an effort to make interaction between human and machine possible. Weizenbaum's ELIZA (1966) became the inspiration for linguistic researchers all over the world in creating a computer program that can understand and respond to the natural language [27]. Nowadays, conversational systems integrate computational linguistics techniques with the widely spread communication over the Internet to interpret and respond to statements made by users in natural language [1]. Brennan et al. describe chatterbots as "artificial constructs that are designed to converse with human beings using natural language as input and output" [2]. The typical usage involves receiving the user utterance in a given human language and providing the possibly reasonable or intelligent response to the given sentence. Then, the sequence is repeated as long as user keeps the conversation going. Since their early days, chatterbots have become more sophisticated along with maturing technology and able to respond to user utterance in both text and synthesized voice speech. Moreover, they evolved from simple text environment into modern visual interfaces that incorporate human-like avatars or virtual presence through augmented reality interfaces [3].

2. CHATTERBOTS: PAST AND PRESENT

Since their origin chatterbots were designed to perform certain functions or, as it could be said, perform certain roles or occupations. ELIZA, simple but revolutionary chatterbot that still influences the research in that field, used simple pattern matching algorithm to find a corresponding response to a given pattern. Despite of its simplicity, this program also performed a role, being a Rogerian psychotherapist able to answer users' questions without any traces of its own personality. ELIZA was later criticized for its lack of an internal world model that could influence and track conversation. Then, a chatterbot named Parry developed by Colby in 1975 simulated paranoid behaviour [4]. By using different tricks like admitting ignorance, changing the level of the conversation, rigidly and continuing previous topics it successfully fooled its human judges into believing that he is one of the patients showing typical paranoid behaviour. That sufficiently explained its unusual responses that would be attributed to illness and make its appearance real. But a real progress in chatterbot technology came in 1995 when Wallace introduced Artificial Linguistic Internet Computer Entity (ALICE)[3]. This chatterbot, often described as modern ELIZA, is an open source project that became the prototype of many current chatterbots. When ELIZA held no memory of the conversation and its knowledge base was embedded right into the code, ALICE introduced Artificial Intelligence Markup Language (AIML) used to store the knowledge-based data. This solution responded to the existing necessity of a bigger knowledge base. The chatterbot created by Dr. Wallace has won Loebner's annual competitions that awards prizes to the most human-like chatterbots, which is the realization of Turing Test.

Exploiting natural language techniques to build conversational systems has been used for a broad range of applications. Nowadays, dialogue systems are deployed on commercial websites to respond to customers' inquiries about provided services. They also answer questions about financial services or verbally demonstrate the company portfolios. Many companies, such as VirtuOz [2], use conversational systems for customer support. They also have a potential to assist the patients as virtual physicians/doctors [5], serve as bully and harassment advisors [6] or act as storytellers [7]. Some chatterbots are created purely for entertainment purposes, deployed in video games to inform players about in-game events [8] or as companions assisting players in a virtual environment [9].

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[2] http://www.virtuoz.com/

[3] http://www.alicebot.org/

Finally, educational conversational systems assist students by providing guidance as they learn [10].

In recent times, chatterbots research has been drawn towards embodied conversational systems where body language and gestures are almost of the same importance as the natural language dialogue [3]. This new approach is motivated by necessity of making conversational systems appear more human like, thus become more engaging with the user. However, when a significant proportion of chatterbots research is concentrating on evaluating such interfaces, evaluating the quality of actual conversation seems to be a research challenge. Dybala et al. [24] mention that in case of task oriented systems it is much easier to achieve user satisfaction because of the very existence of task that requires an interaction between human and system to be completed. In case of non-task oriented dialogue, the interaction depends more on the content of the conversation.

3. TASK AND NON-TASK ORIENTED CLASSIFICATION

Chatterbots are often classified into two conventional groups. The first one often referred to as non-task oriented or free talking, considers the ability of program to chat freely on any given topic, without any obligations, mostly for entertainment purposes. In opposition to non-task oriented systems are task-oriented, specialized systems which put some limitations to system functionalities and usually aim to achieve specific goals. Most previously built conversational systems can engage in either task oriented dialogues to better understand human utterances or non-task oriented dialogues to allow users to enjoy the conversation. To our knowledge, even though there have been some attempts to provide such conversational systems, there are not many existing ones that would successfully employ both functions. A system presented by Nakano et al. [11] can dynamically change dialogue strategy based on speech recognition results. This system features different control modules called experts and each of it is dedicated to perform different kind of task. Non-task oriented functionality is incorporated into this model as one of the modules, the chat expert, which employs standard chatterbot technology (ALICE). This integrated system for both kinds of dialogues can adaptively change the strategies and more accurately respond to human utterances. However, the authors mentioned that it is important to improve chat expert and operate on a larger base of vocabulary – currently limited due to the problems with voice recognition accuracy.

Crocket et al. [12] presented an overview of methodology for constructing goal oriented conversational systems (GO-CA) with a mixed-initiative strategy (from time to time either human or chatterbot may take control of the conversation). By goal-oriented they understand a deep strategic purpose of conversation and directing it to achieve a certain goal, e.g. task oriented dialogues for services and non-task oriented dialogues for chat and entertainment. Authors express the concern that traditional chatterbots prolong the conversation with humans using pointless chat. Therefore, GO-CA attempts to get the conversation back on track using its rule base and knowledge obtained during the conversation. The authors presented two different chatterbots with different goals. The first one called Adam served as University Student Debt Advisor and was focused on providing advice to students. The main requirement of the system was to cope with upset students and deal with abusive language. During the usability evaluation with 200 undergraduate students the subjects were asked to provide their opinion about Adam. 75.0% of students were satisfied by the advice and 47.0% stated that they would use this system instead of going to a real advisor. The majority of student comments were positive, e.g. "(...) I felt like I was seeing a real advisor". Among the areas where Adam could be improved, subject mentioned better understanding of slang and mobile text talk. The current version of Adam has been run at the University and system continues to learn how to relate to life as a student. The latter one was used as an advisor on "bullying and harassment in the workplace" [11]. A complex policy was automated using a conversational system for advice to be available 24 hours a day. The chatterbot allows using natural language to discuss a person's individual cases and offer valuable advice. During the evaluation process 94.0% of responders indicated that they found the advice without difficulty and they didn't feel like consulting the human advisor.

We found these results to be consistent with the research on chatterbots as language learning tools by Fryer et al. [29]. The researchers conducted an experiment with 211 students participating in a sessions with two popular chatterbots ALICE and Jabberwacky[4]. According to the results, most of participants enjoyed using the chatterbots and generally felt more comfortable conversing with the chatterbots than a human language partner or teacher. Such positive foreign language communication experience could enhance students' interest in language learning and improve motivation. 74.0% of participants were asked to write about their experience, and they defined the communication with a system as "funny and entertaining". We can interpret those results with a claim that conversational systems offering both pedagogical functions and a chance of engaging, non-stressful conversation have potential to become artificial teachers. However, the authors did not provide conclusive data and there are still questions whether their results can be treated as reliable. Therefore, more conclusive results should be provided to confirm their claims. Of course a mere chatterbot system that correctly answers the students' questions is not enough to create the artificial representation of a teacher. Among the functionalities that should be considered is to provide conversational agent with a sense of humor and the ability to react to changing emotional states of students, as well as offering some representation of emotions from the side of conversational system in order to make it more human-like. By fulfilling this condition this technology has a potential to offer much more than just mere engaging dialogue.

4. CHATTERBOTS WITH OCCUPATION

From the beginning chatterbots performed different roles supplying counsel and service to others. We claim that just like their human counterparts, conversational systems tend to be

[4] http://www.jabberwacky.com/

specialized in their fields and aim to bring the high standards of professional and intellectual excellence. By the term "occupation" we describe the activity to which one a certain individual devotes oneself that also requires special knowledge or training in a given field. However, performing a specified function does not mean that chatterbot should be strictly task-oriented. In order to develop a system that could meet users' expectations the non-task oriented aspects of chatterbot should also be taken into consideration. We are not proposing a new type of chatterbot systems, but a new way they should be classified. After all, the ability to interact with others by asking a question, commenting and receiving an answer is the foundation of human communication. A mere task oriented conversation seems to be in opposition to that fundamental truth. According to Bogdanovych et al. [13] the purpose of increasing amount of chatterbots in e-commerce is to provide consumers with answers for their questions from an entity that resembles human.

Chatterbots have been widely used in the human-computer interaction and in learning they can potentially serve not only as task-oriented tools, but also as companions that would build students' self-confidence in foreign language communication. Non-task oriented functionality seems to be a great way to allow students a chance to practice their language skills after finishing the main assignment. They would have a chance to review a newly acquired material and have an opportunity to actually use it in practice. Students could also do the self-analysis by seeing the transcript of their conversation and finding/correcting the problematic sentences. A more advanced system could also do it automatically by preparing the summaries of most occurring mistakes and offer guidance how to correct them or dynamically change the teaching strategy to concentrate on filling the gaps in the knowledge of currently studied second language. Also for the teacher it is a chance to track progress of the students and see what language they use and spot the most common mistakes. Finally, by turning text-based conversation into audio they could read and listen at the same time. Creating a chatterbot with task and non-task oriented functions has one more potential advantage – it could be a good language tool for beginners to deal with specific language tasks as well as casual conversation partner for higher level students. As teachers, chatterbots should appear as human as possible to become believable teachers.

So what actually makes the ideal teacher? One of the possible answers has been brought in a detailed report that sets out the ideal educator [14]. According to the simplified definitions provided by surveyed 12-13 years old pupils a good teacher should: listen to students, encourage them, like teaching his/her subject, take time to explain things, help students when they are stuck, allow them to have a chance to speak, not give up on them, care on their opinion, take time to explain thing, be friendly and supportive ("tells you how you are doing"). There are still many possible drawbacks in bringing a full-fledged chatterbot able to serve as language instructor. Coniam [15] in an evaluation of language resources of a few popular chatterbots (including Jabberwacky, ALICE and Jenny[5]) for their potential in teaching English describes a few commonly occurring problems. According to this research, current chatterbots work best when they process one-clause sentences and the topic of conversation is common. Some of them can deal with misspellings but colloquial forms of English cause interaction problems, especially for younger learners accustomed in using this kind of language in everyday conversation. In general, if user inputs incorrectly spelt words, the chatterbot experiences problems or does not understand.

Therefore, a chatterbot that would suggest corrections to certain ungrammatical utterances is necessary for better communication with artificial language tutor. Teenagers often use colloquial and simplified forms of language in their everyday communication, so a successful teacher should be able to recognize such language and, if possible, suggest a correctly spelt form or more appropriate word. Second language students often use different audio-visual materials, i.e. movies, music, games, to improve their language skills. However, they may encounter inappropriate expressions and try to use them during the chatting sessions. Teaching chatterbot should be able to respond this issue in an appropriate way. Recently, different conversational systems have been developed for second language learning [16-17]. The one who has received the most attention is CSIEC (Computer Simulation in Educational Communication) is an Artificial Intelligence (AI) framework developed at Peking University [10]. This system is worth attention because it has been developed over the years and has been successfully integrated into selected Chinese classrooms where it has been used as a language tool to assist teachers during English classes. Even though CSIEC provides learners with a chatting partner, the creators of the system admit that there are still many unsolved problems regarding the ability of understanding and generating natural language or dealing with textual ambiguity.

5. OUR RESEARCH

In our research we aim to create a system that would perform both task and non-task oriented functions to give students not only a chance to expand their language knowledge, but also practice with a virtual language partner who could support their endeavors.

5.1. Baseline chatterbots

Currently, we are using two different baseline chatterbots. First, we have been adapting casual conversation system using modality and word association retrieved from the Web (Modalin), developed by Higuchi et al. [18] to create the baseline of our project. Modalin uses those words association to create propositions and automatically extracts sets of words related to conversation topic selected by user. It also adds the modality to generated utterance and verifies the semantic reliability. It extracts word associations in real time using Goo search engine[6] snippets and does not rely on any off-line database. After applying the extracted word associations into proposition templates, it checks its naturalness on the Internet. If the proposition has low result frequency, it is considered unnatural and system simply omits it to generate a new one in the analogic way.

[5] Jenny, http://juan.vhost.pandorabots.com/botmaster/en/home

[6] Goo search engine, http://www.goo.ne.jp

As a consequence of occurring problems with Modalin system, highly dependent on search engines to answer the user's utterance, we decided to also incorporate the other available chatterbot in our research. For each turn of the dialogue multiple queries has to be made and it leads to overuse of search engine and, in consequence, treating the system as potential spam sender. Therefore, we decided to use the second conversational agent that, unlike Modalin, uses limited amount of queries. A conversational system called Maru-chan [19] extracts keywords from the user's utterance (nouns and adjectives) and uses them to perform fewer queries in the Google search engine. Maru-chan does not only extract world associations from the Internet, but also n-grams, the strings of words containing sets of 3- and 4-grams for every word from the list. Next, these extracted n-grams and candidates are used in order to generate chatterbot response candidates. This system also checks the length of the candidate and automatically deletes the longest and the shortest candidate to deal with sentences that convey too little or too much information. The priority goes to the medium candidates that are scored according to manually set threshold (decided after a preliminary experiment) and the top one are selected as the agent's response.

Meanwhile, we are working on a new version of Modalin that would overcome the occurring problems and we aim to possibly integrate functionalities of both chatterbots to utilize their best features and enrich chatterbot system's conversational strategies.

5.2. System components

The architecture of our system consists of a few independent modules that cooperate with each other to perform the main task (teaching English) as well as the other, supportive functions, e.g. sentence correction, language normalization and responding to users' emotional states in an appropriate way (Fig. 1). The system is built upon the baseline chatterbot (Modalin) with other modules performing a supportive role during the input utterance analysis and generation of output utterance. The functions of each module are briefly described above.

We currently work on implementation of a context learning method based on the phenomenon of code mixing into chatterbot in order to generate code-mixed phrases. This is an innovative way to use this occurrence with conversational agents and to our best knowledge it has not been yet done in the field of computer science. Code mixing is the transition between linguistic units (words or phrases) from one language into another, within one sentence, where original grammar of the native language is usually preserved unchanged [20]. We aim to create the environment where students will be able to learn new language units by context in a natural way, just like children who learn language by connecting words to the people, things, and activities around them. Namely, during conversations with a chatterbot using the base language (Japanese) students will encounter words in second language. By establishing a connection between the meaning of a certain difficult word in the first (L1) and second (L2) language, students may understand the meaning without other references, e.g. dictionaries. For instance: "Let's have a lunch together in the shokudou (cafeteria)" (English-Japanese code-mixed phrase; a real life example).

The results of preliminary experiments described by Mazur et al. [21] indicate that code-mixed sentences may be an effective way of expanding one's L2 vocabulary. Our recent experiment concluded with high percentage of correct answers in a given language understanding test (82.5% and 79.1% for each test phase) also brought and insight that code mixing method may be an effective way of presenting new vocabulary units to the students [22]. Code mixing module will let users engage in both task and non-task oriented conversation. In a free conversation mode this system will automatically generate Japanese sentences with English words and allow users to elaborate on their meaning while conversing with a system.. The non-task oriented functionality will allow users to select vocabulary from a given list (currently nouns) and system will use them to generate sentences using Conceptnet[7], a free common sense knowledge base. Users will encounter both real commonsense sentences and fake sentences that do not convey the meaning of a studied word. In a sentence generation we also would like to use Wordnet[8] to check the semantic distance between response candidates and eliminate the ones which are too close and may interfere with our context learning approach.

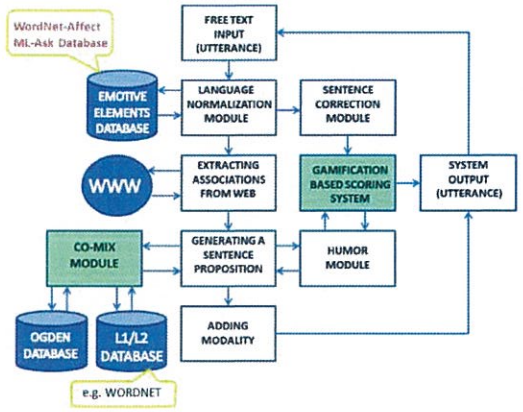


Fig. 1 System overview

Next, we plan to make use of an emotion recognition module that will recognize emotive sentences with ML-Ask Emotive Elements/Emotive Expressions Analysis System for Japanese, developed by Ptaszynski et al. [23]. This system determines utterances' emotiveness and detects types of emotions. The two main features are determining general emotiveness of a sentence (emotive/non emotive) and specifying the types of emotions. For example, the sentence "Kyo wan nante kimochi ii hi nanda! (Today is such a nice day!)!" is recognized as emotive, and contains emotive expression ("kimochi ii (nice)") as well as emotive elements ("nanda (emphasis)") and exclamation mark that make the utterance more emotive. In the second step, system checks the emotive values of the utterance and conduct the analysis of specific types of emotions. For example, the above example sentence contains an emotive expression "kimochi ii

[7] Conceptnet, http://web.media.mit.edu/~hugo/conceptnet/
[8] Wordnet, http://nlpwww.nict.go.jp/wn-ja/

(nice)" which belongs to the group called "yorokobi (joy)". This group is considered as positive, in opposition to negative emotion group, such as "iya (dislike)". Emotion recognition system will help us deal with shifting attitude of users towards the system, as well as the problem of keeping user engaged and eager to continue the discussion with conversational system.

The other aspect we take into consideration is humor that can be used to enhance positive and reduce negative engagement. We intend on using pun generation system made by Dybala et al. [24]. This system extracts a base word from a user utterance (mostly nouns) and transforms it using Japanese pun phonetic generation patterns to create candidate list. After checking the candidates in the Goo search engine and selecting the one with the highest hit rate it looks for a sentence with a chosen word and extracts its part to form a response or randomly selects a pun from its pun database. Dybala et al. also conducted a study on the role of engagement in non-task oriented conversation and the influence of humor as a measure to improve the engagement. The authors underline the fact that there is no comparative study concerning the role of engagement in conversation with task and non-task oriented systems. Finally, we would like to aid non-native speakers' reading comprehension of informal English with language normalization module based on a CEGS, a system for generating casual English short sentences from regular English input using a phonetic rule-based approach, developed by Clark et al. [25].

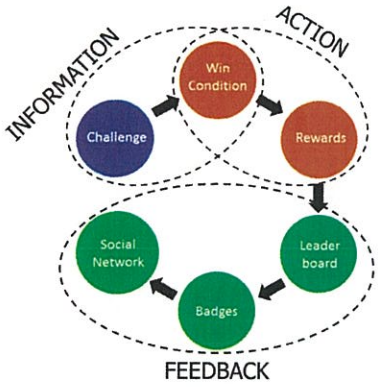


Fig. 2 Gamified point system

The other way we would like to deal with user willingness to enforce our system with a gamification module [26], dedicated to encourage user engagement and increase willingness to interact with a chatterbot. Gamification is an informal term for the use of elements of video game design in non-gaming systems. It is an innovative way of designing modern applications highly inspired by electronic entertainment. The chatterbot-based nature of our system requires other ways to encourage user engagement. Some of existing works on dialogue systems, especially the one by Jia at al. [10] show that the quality of conversation with chatterbot can be measured by its duration and general willingness of the user to interact with a system. According to this research, without a specific goal of the conversation user engagement slowly decreases. Since our initial target group consists of high school students, we decided to provide a scoring system inspired by gamification ideas, with a sole purpose to motivate users to learn foreign language and reward their efforts. Most of the schools use grades to motivate students and control their progress – high grades for excellent results, and low grades as a consequence of the lack of studying. There has been some controversy about whether grading students is an appropriate practice, because it may actively discourage them. We, on the other hand, believe that modern teaching system should offer more than just mere grading system. Using gamification ideas may bring the motivation and positive attitude of users towards the system. Therefore, such mechanism may encourage chatting with an agent and serve as a mean to keep user engaged and eager to study at longer intervals of time.

Currently used scoring systems, like the one presented by Jia et al., seem to be limited with their progress assessment methods and founded on simple grading rules, thereby allowing users to review their performance and scores only in a very basic way. Nevertheless, this work proves that such function is important to self-learning and evaluation. Therefore, we would like to introduce a more recent approach to this matter with a gamified point system that not only assess student progress, but also rewards them for their achievements and encourage doing more effort in studying. This point system is an iteration starting with a presenting a challenge for students (Fig. 2). Firstly, we explain rules of the game: present the challenge (e.g. acquiring and actively using newly learnt vocabulary units), specify winning conditions and rewards for reaching certain goals. Among the possible gratifications of user efforts the elements such as leaderboard and badges - commonly used game mechanics - will provide motivating competitiveness to users. Leaderboards show the top rank students and their best scores. Just like in video games, the urge to beat the best score and place one's own name among the top rank users has a positive motivation quality. Badges are commonly used in games and also effectively serve its purpose many popular social network applications, such as Foursquare[9]

Finally, the popularity of social media networks will give the opportunity to reward users with recognition of their achievement in the web community among their friends and relatives. A reward, such as dedicated badge, can be obtained automatically, after completing the necessary set of tasks.

6. CONCLUSIONS

This paper presented the insight on the roles the conversational systems play to benefit the modern society. We underlined the fact that existing informal classification of chatterbots is not sufficient enough to express the growing expectations of potential consumers. In a discussion about chatterbot occupation we emphasize the possible adaptation of conversational agents in learning. As an example, we described our work on an English tutoring system that presents a new approach to the subject by taking into consideration both aspects of task and non-task oriented conversation to face the student expectations towards a

[9] https://foursquare.com/

successful artificial teacher. We presented some examples showing that such system should perform both task and non-task oriented functions to become a useful language-learning tool and properly perform its designated role. We also discussed some existing problems in currently used chatterbots and presented some possible solutions to solve these issues. The originality of our project lays comes from the idea of using different modules performing separate functions, e.g. a module that use code mixing-based context method to generate sentences used in conversation with a chatterbot; gamification-inspired scoring system to encourage user engagement; a humor module that generates simple language jokes to deal with changing user attitude towards system and motivate; a language normalization module to improver comprehension of informal English; finally a module for affect analysis to generate and analyze emotive sentences.

A successful chatterbot must offer more than just performing its designated tasks, because potential users are expecting more human-like interlocutors that will not only answer their questions but also allow them to have a free conversation. Therefore, designing a new chatterbot should take into consideration both aspects of conversational systems and create a bridge between them in order to fulfil the expectations of potential users.

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